SOFTWARE REQUIREMENTS SPECIFICATION

AI-Based System-Wide Real-Time Masking of Offensive Language and People on Android Phones

An Android app that hides bad language and images of people on the screen, in the apps people use every day (limits in §5).

Version	Date	Scope
2.3 (draft)	18 October 2026	Android: v1–v4 graduation, v5 bonus, v6 future work

Status

For team review

1. Introduction 2. Overview 3. Features 4. Quality targets 5. Not planned 6. Status & plan 7. Risks 8. Terms

1. Introduction

1.1 Purpose

This document explains what the app must do and how well it must do it. It is written for the project team and supervisors, so it avoids deep technical detail. The full technical study and the test results live in `investigation-android-v1.md` and `p0-results-android-v1.md`.

1.2 The problem

Some people want to avoid bad language and images of people while they use their phone: in comments, chats, social apps, videos, ads and news. Today's tools have three problems:

- Most work only in **one place**, such as one browser or one website.
- Tools that watch the whole screen show the content **first** and cover it **after**, so it flashes for a moment.

- Some send screen content to a server, which is bad for privacy.

1.3 Our idea

The project is one Android app. It uses Android's accessibility feature (the same one screen readers use) to know what is on the screen, and it draws covers on top. Everything runs on the phone.

- **Bad language:** the app reads the text that apps show, finds bad words with a word list, and covers them. Our own small AI model on the phone adds context: it catches insults without bad words and avoids covering normal uses of words such as "dog".
- **People:** the app looks at a picture of the screen and finds people with an AI model on the phone, then covers them. Videos while they play are future work (v6, after graduation).
- **Text inside images (v2):** the app reads the text inside pictures (OCR), for example memes and thumbnails, and covers the bad words.

1.4 Release plan

VERSION	HIDES	WHY IN THIS ORDER
v1	**Bad-language text** in standard Arabic and English: swearing, sexual words, insults, blasphemy. A word list plus our own small AI model, which adds context (the team's main research contribution).	The easiest start. It builds the app that later versions reuse.
v2	**Text inside images** (memes, thumbnails, subtitles burned into videos)	Builds the window-picture step that the people versions reuse
v3	**Big people** (taller than about 96 screen pixels)	The AI model already finds most of them
v4	**Medium people** (about 32–96 pixels)	Needs a closer check in parts of the screen
v5 (bonus)	**Small people** (under about 32 pixels)	Needs our own training on phone screenshots

`v6 (after graduation)`	**Video**: hide people in videos while they play	The hardest: needs more checks than 3 pictures per second

Each version adds to the one before. **Graduation scope: v1 to v4.** v5 is a bonus if time allows; v6 is future work after graduation. (Older documents call the text model "v1.1"; it is now part of v1.)

2. Overview

2.1 How it works, simply

Bad language (v1):

- **Listen:** Android tells the app when the screen changes.
- **Read:** the app reads the text on the screen.
- **Clean:** it removes spelling tricks (for example tashkeel, stretched letters, symbols in place of letters).
- **Match:** it looks each word up in the bad-word list.
- **Cover:** it puts a solid block over the bad word or over the whole message (the user chooses).

Text inside images (v2): the app takes a picture of the other app's window, the OCR model reads the text in it, the word list and our model check that text, and bad words are covered.

People (v3–v5): the changed part of the screen is covered at once. The app takes a picture of the other app's window (the picture does not include our covers), the AI model finds the people, and the cover is removed everywhere except over the people.

Content changes	Covered while checked	Shown
scroll, new page, new app	people: about 0.3–0.7 s today	with every detected person hidden

Text needs no "covered while checked" step. The check takes less than a millisecond, so the cover appears right after the text.

2.2 Users

One kind of user: a person who chooses to install the filter on their own phone. It is a personal tool, not a parental-control or monitoring tool.

2.3 Where it runs

ITEM	REQUIREMENT
System	v1 (text): Android 8.0 or newer. v2–v5 (everything that needs a picture of the screen): Android 14 or newer. Older phones are not supported for these versions.
Test phone	Xiaomi Redmi 12: MediaTek Helio G88, 2 fast + 6 slow processor cores, small Mali-G52 graphics. A normal low-cost phone.
Internet	Not needed. The app does not even ask for internet access.
Install	From an APK file during development. Google Play later: the team already has a Google Play developer account.

2.4 Assumptions

- The user turns on the app in the phone's accessibility settings and allows it to run in the background.
- "Bad language" means the words and phrases on our list, in four categories: swearing, sexual words, insults and blasphemy.
- "Person" means any human figure: real, drawn, animated or in a game.
- Hiding something by mistake is acceptable. Showing a person or a bad word by mistake is not. When in doubt, hide.

3. Features (functional requirements)

Priority: **Must** = needed for that version, **Should** = planned if time allows, **Could** = later.

3.1 The app (built in v1, used by all versions)

ID	REQUIREMENT	VER.	PRIORITY	STATUS
FR-1	Before turning on, show a clear screen that explains what the app can see and that nothing leaves the phone.	v1	Must	Planned
FR-2	Show a small notification while the filter is on. If the filter stops or crashes, the notification goes away and	v1	Must	Planned

	the user can see that.			
FR-3	Covers move with the page when the user scrolls, so covered content stays covered and the cover does not flicker.	v1	Must	Planned
FR-4	Let the user turn the filter on and off and skip chosen apps.	v1	Should	Planned

3.2 Bad-language text (v1)

ID	REQUIREMENT	VER.	PRIORITY	STATUS
FR-5	Read the text shown by every app that shares its text with Android: social apps, chats, browsers, news.	v1	Must	Planned
FR-6	Find bad words and phrases in standard Arabic and English, in four categories: swearing, sexual words, insults, blasphemy.	v1	Must	List needed The Arabic list must be built by the team.
FR-7	Catch common spelling tricks: tashkeel, stretched letters (ـ), different alef and ya forms, repeated letters, symbols in place of letters.	v1	Must	Planned
FR-8	Match whole words only, so a normal word that contains a bad word is not covered.	v1	Must	Planned
FR-9	Cover with a solid block. The user chooses: cover only the bad word, or the whole message.	v1	Must	Planned
FR-10	Let the user turn each category on and off.	v1	Should	Planned
FR-11	Do not cover text the user is typing.	v1	Should	Planned
FR-20	Our own small on-phone AI text model as a second layer: it reads the whole message, decides the category with a confidence, and points at the offensive **words**. It handles context-dependent words and insults without bad words. Trained by the team from large open "teacher" models (investigation §3.5). Compared with the	v1	Should	Planned

	word list and ready-made small models.			
FR-21	The user chooses which language(s) to filter: **one by default** (the phone's language), two if needed. Each language is a "pack" with its own word list and its own model. Only the chosen packs use memory. Text is sent to the right pack by its letters (Arabic or Latin).	v1	Must	Planned

3.3 Text inside images (v2)

ID	REQUIREMENT	VER.	PRIORITY	STATUS
FR-12	Read text inside images and videos (memes, thumbnails, subtitles burned into videos) from the window picture, and cover the bad words in it. Planned with PaddleOCR (Arabic mobile model, about 8 MB). Google's on-device OCR does not support Arabic.	v2	Must	Needs a phone test OCR speed on the phone is not measured yet (step P0d).

3.4 People (v3–v5)

ID	REQUIREMENT	VER.	PRIORITY	STATUS
FR-13	Detect people in every app that shows images and allows screenshots. Cover each one with a strong pixelation or solid fill, a little larger than the person. A light blur still shows the body shape.	v3	Must	Tested offline
FR-14	Cover changed content as soon as the phone reports the change, and keep it covered until it has been checked. The cover never lifts by itself.	v3	Must	Planned
FR-15	Apps that block screenshots (banking, Netflix and other protected video apps) cannot be pictured, so people and text inside their images are not checked there. Their normal screen text is still filtered. An Android limit that no normal app can avoid. Optional: an "always cover this app" switch, which covers the whole window.	v3	Could	Platform limit

FR-16	Fallback for Android 13 and older: screen capture instead of window pictures. Dropped for now (18 Oct 2026): the app is built for Android 14+ first.	—	Not now	Dropped
FR-17	Do a closer check in parts of the screen (tiles) to find medium-size people.	v4	Must	Needs work Works offline. Too slow on the phone today.
FR-18	Find small and far-away people, using a model trained on real phone screenshots.	v5	Bonus	Needs work

3.5 Video (v6, after graduation)

ID	REQUIREMENT	VER.	PRIORITY	STATUS
FR-19	Hide people in videos while they play (full screen and inside feeds), often enough to follow movement. The window picture allows at most 3 pictures per second, so v6 needs its own method: tracking people between checks, or a faster picture source for video only.	v6	Future	Needs research

4. Quality targets (non-functional requirements)

ID	AREA	TARGET	TODAY
NFR-1	Text safety (v1)	Hide at least 95% of the bad words on our list in real comments and chats.	Not measured
NFR-2	Text false covers (v1)	No more than 1 wrong cover per 1 000 clean words (news, Quran, normal chats).	Not measured
NFR-3	Gap before the cover	New content shows for no more than about 3 frames (about 33 ms on a 90 Hz screen) before it is covered.	Not measured First measurement in v1.
NFR-4	People safety (v3–v5)	Hide at least 95% of people of that version's size: big in v3, medium in v4, small in v5.	Not yet On test screens, one pass (with tiles): big 85–89% (91–95%), medium 60–

			76% (87–89%), small 24–27% (51–54%).
NFR-5	Waiting time (people)	Changed content stays covered for no more than about 0.7 s.	**Not measured** Expected 0.3–0.7 s from the parts.
NFR-6	Check speed (people)	One full-screen check takes 150 ms or less on the test phone.	**Not yet** GPU about 205 ms (189 ms over 15 minutes non-stop); best processor setting 277 ms. Measured 7 Oct 2026.
NFR-7	Smoothness	Touches pass through the covers. The phone does not feel slower.	**By design**
NFR-8	Battery	No work while the screen is still.	**By design** Battery use not measured yet.
NFR-9	Memory	v1 under 50 MB. With the text model under 150 MB. With the people model too, under 250 MB.	**Likely** People model alone: about 87 MB (processor) to 180 MB (GPU) at peak.
NFR-10	Privacy	Screen text and images stay in memory only. Nothing is saved and nothing is sent. The app has no internet permission.	**By design**
NFR-11	Stability	A crash in the app must never close or freeze other apps.	**By design**
NFR-12	Permissions	The user grants access **once**. The app never asks again on every use or after a restart. A common complaint about a competing app.	**By design** v1 needs no screen capture. v2+ uses the accessibility window picture, which needs no per-session prompt.
NFR-13	Text model	Per language pack, on the test phone: real RAM at most 100 MB, at most 50 ms per message (p95), about one CPU core at most while checking (or the GPU). Better than the word list alone and than ready-made small models on the same test set.	**Not built** Candidates are measured on the phone first (step P0c).

NFR-14	Text in images (v2)	Targets for hidden words and speed are set after the OCR phone test (P0d).	Not set

5. Not planned

- iPhone. Apple does not let apps see or cover other apps.
- Arabic dialects and Arabizi (Arabic written in Latin letters). Possible after v1.
- A root mode that works inside Android's screen drawing
- The Linux desktop version. It is paused as plan B, and its test results still help.
- Hiding only some people, for example by gender. This is much less accurate, and a mistake means a person is shown.
- Blocking the user from turning the filter off. It is a personal choice, not a lock.
- Pictures of apps that block screenshots (banking, Netflix and other protected video apps): Android does not allow it. Their screen text is still filtered.
- Text in apps that do not share it with Android (for example many games). Only the picture-based checks (v2–v4) work there.
- Video while it plays (v6): future work after graduation.
- Phones older than Android 14 for v2–v5, for now.

6. Status and plan

STEP	WHAT	TIME	STATUS
v1-a	App skeleton: accessibility service, covers, settings, notification, consent screen	~1 week	Next
v1-b	Text filter: clean, match, locate the word, cover, follow scrolling. Measure the gap.	1–2 weeks	Planned
v1-c	Word lists: build the Arabic list, review the English list, test on real comments	~1 week	Planned
v1 model	Own text model: data, labels from large open models, training, comparison	4–5 weeks	Planned

P0	Test the people model for speed and accuracy on a laptop and on test phone screens	days	Done
P0b	Finish the speed test on the Redmi 12	1 evening	Done GPU about 205 ms per screen
P0d	Test Arabic OCR (PaddleOCR) for speed and accuracy on the Redmi 12	days	Planned
v2	Window picture, OCR, cover bad words inside images	3–4 weeks	Planned
v3	People model in the app, big people	~2 weeks	Planned
v4	Tiles and a faster model setup, medium people	1–2 weeks	Planned
v5	Train on real phone screenshots, small people	weeks	Bonus

What the tests showed so far: the people model finds big people well. Small people and phone speed are the open problems, and they come later (v4, v5). v1 does not depend on them.

7. Risks

RISK	EFFECT	WHAT WE DO
No complete free Arabic word list (the best free one has 348 Arabic-script entries)	Many Arabic bad words are missed	Start from free lists (LDNOOBW V2, Attia's lexicon). The team completes the list, then tests it on real Arabic comments.
Insults and blasphemy depend on context	Normal text gets covered	Phrases instead of single words, a strict insult list, a switch per category

Some apps do not share their text	Bad words in those apps are not covered	List the apps that work. Text from pixels (OCR) in v2.
Small people are missed	Some people are shown	Tiles, training on phone screenshots, hide when unsure
The phone is too slow	Long waiting time, no tiles	Faster model types, graphics chip, smaller input. Otherwise a stronger test phone.
Xiaomi's battery saver stops the app	The filter stops without warning	A setup screen that asks for Autostart and no battery limits. The notification (FR-2) shows when it stops.
"HaramBlur" already exists as a browser extension and an Android app (about 700K installs)	Our project must offer something it does not	Bad-language text, text inside images, cover only the person, hide first without a timeout, most apps. Cite it as related work. Our old working name was too close to it, so it was dropped.
Arabic OCR is too slow or too weak on the phone	v2 waits long or misses words	Measure first (P0d). Check only images that changed. Smaller input.
Video needs more checks than the phone can do	People in videos show between checks	Track people between checks. In strict mode, keep the video covered.

8. Terms

Frame	One picture on the screen. The Redmi 12 shows up to 90 per second, so one frame is about 11 ms.
ms / s	Millisecond (one thousandth of a second) / second
Accessibility service	A special kind of Android app that the user turns on in settings. It can read what is on the screen and draw on top of it. Screen readers use the same feature.
Cover	The solid block or strong pixelation that we draw over a bad word or a person
Word list	Our list of bad words and phrases, each tagged with a category

Cleaning (normalization)	Removing spelling tricks before matching, so "كلبيبب" and "كلب" count as the same word
OCR	Reading text from a picture. Used for text inside images in v2 (PaddleOCR).
AI model	A trained program. Ours find offensive text, read text in pictures, or find people.
Tiles	Checking the screen in a few parts instead of all at once, so small people look bigger to the model
Hide first	Our rule for people: new content stays covered until it is checked (in technical documents: "fail-closed")
